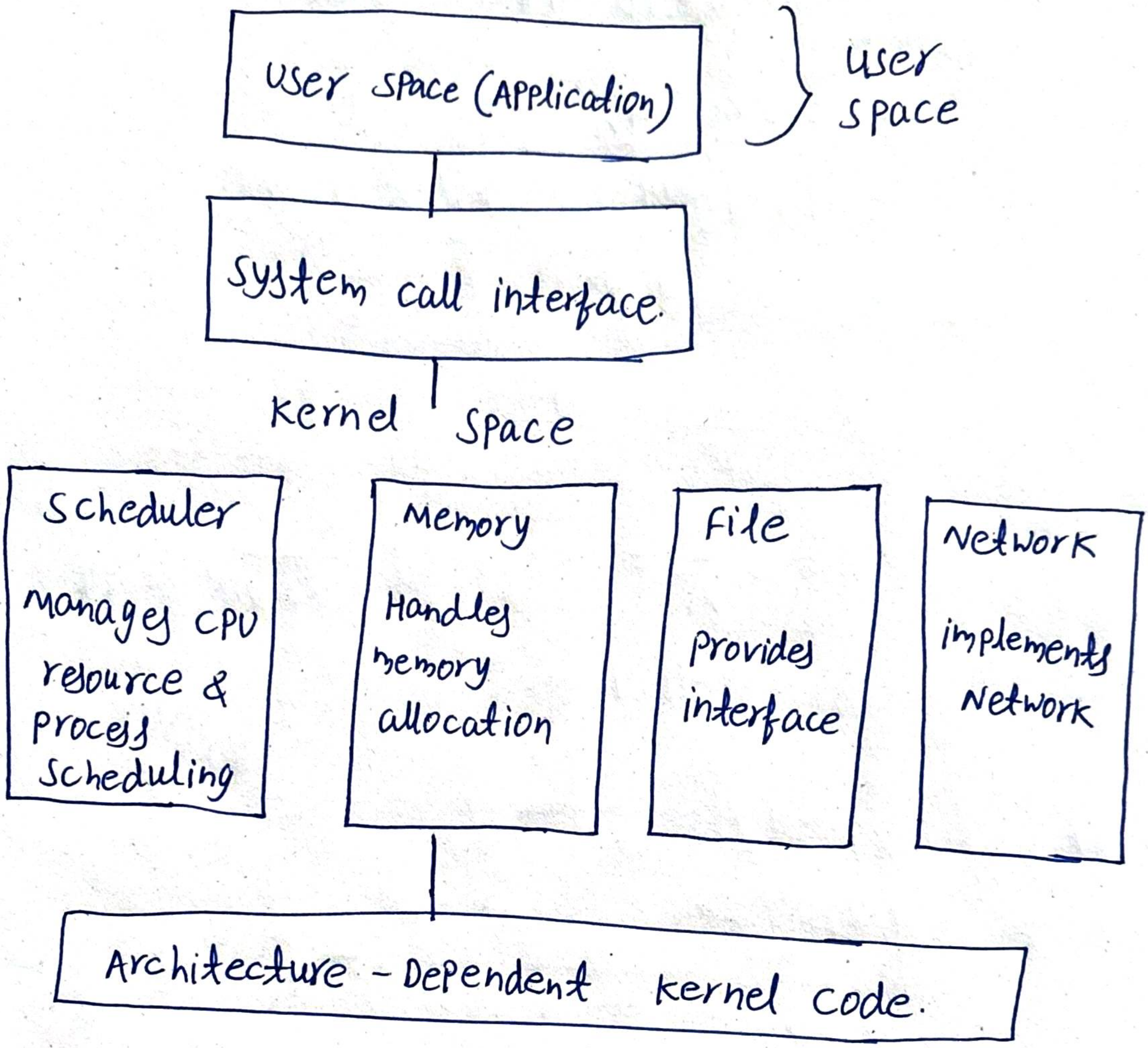
LSP TASK-1

Linux Kernel Architecture



Scheduler: manages CPU time allocation to multiple Processes and threads. It ensures fair CPU usage and handles Process Priorities and multitasking.

Memory Management: controls system memory allocation, including virtual memory, paging, and caching. it ensures safe memory sharing between Processes.

File system: Handles how data is stored and retrieved. It uses the virtual file system (VFS) layer to support multiple file system like NFS, FAT32, etc.

Network stack: Manages all network-related operations, including communication over protocols like TCP/IP, sockets, and packet routing.

Device drivers: Provide interfaces between the kernel and hardware devices. Each driver translates kernel instructions into device-specific operations.

System call interface (SCI): Acts as a bridge between user application and kernek functions.

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